MUPIROCIN

MRSA decolonisation

Max 10 days

CHEMICAL NAME 9-[(E)-4-[(2S,3R,4R,5S)-3,4-dihydroxy-5-[[(2S,3S)-3-[(2S,3S)-3-hydroxybutan-2-yl]oxiran-2-yl]methyl]oxan-2-yl]-3-methylbut-2-enoyl]oxynonanoic acid (pseudomonic acid A)

CLASS Topical antibacterial (monoxycarbolic acid)

TRADE NAMES Bactroban · Bactroban Nasal · Supirocin

THERAPEUTIC USES / INDICATIONS

- Nasal decolonisation of MRSA (and *Staph. aureus* before surgery/dialysis).
- Impetigo (localised) and infected small wounds, folliculitis.

ADMINISTRATION

- **Nasal ointment:** match-head amount inside each nostril, then press nostrils together to spread — 2–3 times daily for 5 days.
- **Skin:** apply to affected area up to 3 times daily; may cover with dressing.
- Usually combined with chlorhexidine body wash for decolonisation.

DOSAGES

- **MRSA decolonisation:** nasal ointment 2% three times daily for 5 days.
- **Impetigo:** 2% ointment/cream three times daily for up to 10 days.

PHARMACOKINETICS

- Minimal systemic absorption; any absorbed drug rapidly metabolised to inactive monic acid.

PHARMACODYNAMICS

- Inhibits bacterial **isoleucyl-tRNA synthetase** → blocks protein synthesis → bacteriostatic/bactericidal against staphylococci (incl. MRSA) and streptococci.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Local burning, stinging, itching.
- Nasal: rhinitis, taste disturbance.

ADVERSE EFFECTS (SERIOUS)

- Contact dermatitis.
- Resistance with prolonged/repeated use.
- Polyethylene glycol base: nephrotoxicity if applied to large open wounds in renal impairment.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Use on large areas of broken skin in renal impairment (PEG base).

KEY INTERACTIONS

- No significant interactions.

NURSING MANAGEMENT

- Follow MRSA screening/decolonisation protocol; document courses.
- Avoid repeated or prolonged courses (resistance).
- Infection control precautions.

PATIENT EDUCATION

- Use exactly for 5 days (nose) or up to 10 days (skin), not longer.
- Wash hands before and after application.
- Use the chlorhexidine body wash as directed.

KEY POINTS

- Topical antistaphylococcal (incl. MRSA).
- Impetigo: ≤10 days.
- Nasal decolonisation: TDS × 5 days.
- Limit use; resistance.